2021 MALAWI SCHOOL CERTIFICATE

1.

a. Describe any one component of a scientific investigatio

n. ____
____
____ **(1 mark)**

b. State any two importance of drawing line graph to interpret data in a table during scientific investigatio

n.

i. ____
____ **(1 mark) i**

i. ____
____ **(1 mark)**

c. Two elements S and T form ions with the electron configuration as shown belo

w. ? - ? + : 2,8,8

i. Write down the electron arrangement of the atoms of S and T . S
____ **(1mark) T**
____ **(1 mark)**

d. Give any two physical properties of element T .

i. ____ **(1 mark) i**

i. ____ **(1 mark)**

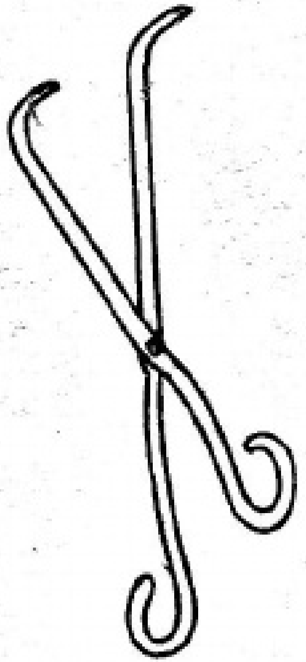
2.

a. Explain the effect on atomic radius when valence electrons are los

t. ____
____
____ **(2 marks)**

b.

Figure 1 below shows laboratory apparatus (i) and (ii)



(i)

i. Write a balanced chemical equation for the reaction between zinc (Zn) and

sulphate (

____ (2 marks)

ii. Use the results to arrange the metals in order of increasing reactivity

____ Page 6 of 14 i

i. Work out the oxidation number of sulphur (S) in a sulphite ion (SO 3 ?-) given that the oxidation number of oxygen (O) is ?

2. (2 marks 5 .

a. Figure 2 shows structures of some organic compounds A, B, C, D and

E. | H CH 3

i. Name compound

A. ____ (1 mark) i

i. Which compound is soluble in water? ____ (1 mark) ii

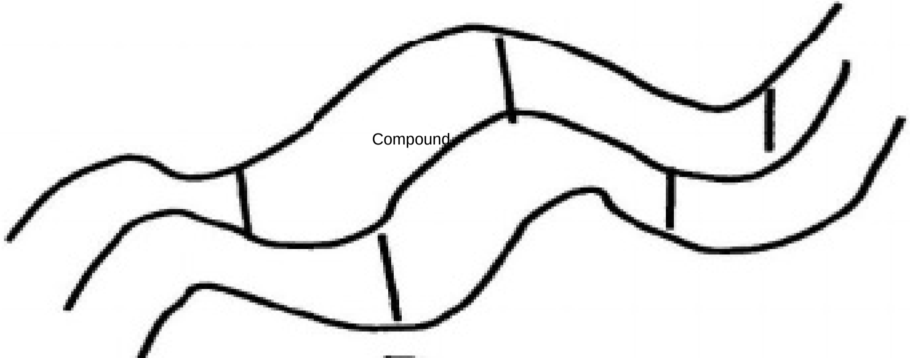
i. Give a reason for the answer in

5.

a. i

i. ____
____ (1 mark) C H C C C C H H H H H H H C H H C H H C H C H H C O OH C H H C C C H H H H H H H A B C D E

A



B

iii.

6. Table 2 gives readings of a

as an unknown substance was heated from solid state to gaseous

b. Use the graph to predict if the unknown substance was pure or impur

e. ____ **(1 mark)**

c. What was the temperature of the substance after 28 seconds?
____ **(1 mark)**

d. Why is it necessary to specify the temperature of a substance when investigating its purity using density?

____

____ **(1 mark)**

7.

a. What is a 'limiting reagent' in a chemical reaction?

____

____

____ **(1 mark)**

b. Calcium carbonate can be decomposed by heating according to the following equation: Ca CO 3 + CO ? (g) on heating 25.9g of Calcium carbonate (Ca CO 3) ? (4 marks)

8.

a. Name the two allotropes of phosphoru

s. ____ **(2 marks)**

b. Give any two differences between diamond and graphit

e.

i. ____
____ **(1 mark) i**

i. ____
____ **(1 mark)**

c. What name is given to the process of manufacturing ammonia gas from nitrogen?
____ **(1 mark)**

e. How is nitrogen useful in transporting petroleum products?
____
____
____ **(1 marks)**

f. State any two uses of sulphu

r.

i. ____
____ **(1 mark) i**

i. ____
____ **(1 mark)**

9.

a. What is a "weak acid?" ____
____
____ **(1 mark)**

b. Mention the acid which could be used to prepare ammonium chloride (NH 4 Cl).
____ **(1 mark)**

c.

i. Reacting an acid with a metal

ii. Reacting an acid with a carbonate

1

1.

a. Explain any one physical method of removing water hardnes

s. ____
____
____
____
____ **(2 marks)**

b. With the aid of well labelled diagram explain how water circulates continuously between the earth's surface and the atmospher

e. ____
____
____
____
____
____
____
____
____
____
____ **(8 marks)**

____ Page 13 of 14

1

2. With the aid of a well labelled diagram, describe how alcohol can be produced locally from cereals, sugar and wate r. ____
____
____
____
____
____
____
____
____
____
____
____
____
____
____
____
____ **(8 marks)**

1

3. Describe how 250cm 3 of a 0.5M magnesium sulphate solution could be prepared using hydrated magnesium sulphate crystals (MgS O 4 ? O) . ____________________

____________________ **(10 marks) END OF QUESTION PAPER**